First Biweekly Report

During this biweekly period, the EduAI project completed its initial setup and partial frontend and backend development. The team invested approximately 70 hours in total, advancing four core areas in parallel: data preparation, frontend prototyping, backend infrastructure, and AI feature integration.

Key Achievements Review

Course Data & Knowledge Preparation:

Completed the initial collection and screening of teaching materials for the first batch of target courses, including Data Structures and Computer System Architecture. The team not only compiled course PPTs and reference materials but also preliminarily constructed a hierarchical system of knowledge points (e.g., linear structures, instruction systems, storage systems).

Frontend Prototyping:

Completed the local implementation of major modules, including Login, Dashboard, Intelligent Q&A, Learning Analytics, and Agent. Currently, each page operates independently using mock data, validating the interaction flow, UI structure, and the logic paths for switching between teacher and student roles.

Backend Infrastructure & Environment Validation:

Comprehensively validated the operation of technology stacks, including PostgreSQL. Furthermore, by refining environment-check scripts and dependency definitions, the team ensured highly efficient synchronization. Concurrently, the team further improved core database models and executed a seed data script to initialize test accounts, 3 foundational courses, and enrollment data, as well as constructing a knowledge point relation graph for the Java course.

Early Feature Integration & Closed Loop:

Successfully implemented a preliminary closed loop for the course resource processing chain. A complete resource processing pipeline was built: it supports secure file uploads to MinIO; employs a combination of fixed-size and paragraph-based chunking for efficient text splitting; and calls APIs to generate embedding vectors for database storage. Uploaded files are automatically saved to MinIO and trigger these asynchronous processing tasks. Meanwhile, the backend conversation flow has been successfully connected to real LLM APIs.

Summary and Next Steps

The project has established a solid foundation of frontend and backend technical support. In the next stage, the team will focus on building a Q&A MVP (Minimum Viable Product) to achieve intelligent question-answering with resource-aware capabilities.